

UTKARSH MISHRA

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Education

Indian Institute of Technology, Delhi (IIT Delhi)

2023 – 2025

M.Sc. in Cognitive Science

GPA: 7.5/10

Relevant Coursework: Advanced Data Analysis for Behavioural Research using R, Advanced Experimental Methods, Advanced Computational Methods, Research Methods, Scientific Writing, Design for User Experience, Design for Usability.

Jamia Millia Islamia, New Delhi

2020 – 2023

B.A. in Economics and Psychology

GPA: 8.33/10

Relevant Coursework: Microeconomics, Macroeconomics, Money, International Trade, Indian Economy, Theories in Public Administration, Indian Public Administration

Research Experience

Research Associate, IIM Ahmedabad

Project 1: Ex-Ante Impact Assessment (Public and Behavioural Perception), India's Olympic Bid 2036 *In collaboration with EY and Gujarat Sports Infrastructure Development (GSID), Govt of Gujarat. Collaborating PIs: Prof. Jeevant Rampal, Prof. Anindya, Prof. Anish, Prof. Amit*

- Designed and administered a **large-scale perception** study of approximately 6,000 respondents to characterize **public attitudes and perceptions** surrounding India's proposed Olympic bid.
- Managed survey design, Qualtrics implementation, data collection and quantitative analysis, combining structured responses with open-ended feedback to identify population-level patterns.
- Applied **sentiment analysis** in R and Python to identify recurring themes and characterize positive and negative public perceptions across the respondent pool.
- Synthesized large-scale **behavioral and attitudinal data** into actionable insights for project stakeholders, distinguishing common perceptions from emerging themes and areas of importance.

Project 2: Impact Evaluation of Rural Development Programme (MoRD, Government of India) *Collaborating PIs: Prof. Jeevant Rampal, Prof. Anindya*

- Contributed to the **end-to-end** design of a **mixed-methods** impact evaluation of a Government of India rural development scheme; defined the outcome indicators, study design, and **sampling strategy**, alongside structured **stakeholder consultations** across administrative levels.
- Designed and implemented **beneficiary and cadre surveys** through literature review, implementation analysis, and stakeholder inputs; coded the survey instrument in (**Qualtrics**), translated, and **piloted** instruments with iterative refinement.
- Coordinated **large-scale primary data collection** across **three states** ($n > 6,000$); **hired, trained**, and managed survey teams of over 30 members in each district for field operations.
- Conducted data cleaning, organisation and **data analysis** in R and Python, including regression, balance checks, and **sentiment analysis**; constructed composite outcome indices.
- Conducted **field interviews** for case study development using thematic analysis; **co-authored** the final report and contributed to presentations to MoRD, informing programme revisions.

Research Assistant, Department of Design, IIT Delhi

Psychometric Instrument Development, Supervised by Prof. Jyoti

- Designed a **quantitative scale** to reliably measure the subjective **experience of enjoyment** in **children** aged 3–6 during indulgence in games and toys meant for educational purposes.
- Conducted **expert panel review** and **cognitive interviewing** to refine item wording and ensure comprehension across diverse respondent backgrounds.

- Administered the scale on a sample of **130 participants** in Delhi; statistically validated the instrument using **Exploratory Factor Analysis (EFA)**, **Confirmatory Factor Analysis (CFA)**, and **internal consistency measures** (Cronbach's α), producing a validated quantitative tool.

Research Project

Impact of AI Placebo Effect on Task Performance in Human Computer Interaction

- Investigated the impact of **perceived AI-based assistance** on user experience and task performance in **Human-AI Interaction** settings by reanalysing the data of an existing published study.
- Implemented Bayesian statistical modelling and Bayesian Inference in Python to identify **latent psychological factors** driving variation in task performance, recovering individual-level parameter estimates and uncertainty quantification.
- The findings got accepted for presentation at the Annual Conference of Cognitive Science, 2025.

Master's Thesis, IIT Delhi

Optimising Human-AI Interaction in Digital Learning Supervised by Prof. Jyoti, Head, Dept. of Design

- Investigated how **AI systems** can help improve **learning outcomes** in **digital learning setups** by identifying the type of content that is easily readable and generating that content using **LLMs**
- Implemented an attention-based **multimodal AI** integrating **text and audio features** for **emotion recognition** in conversational learning contexts; addressed data imbalance with ensemble learning, significantly improving recognition of under-represented emotional states.
- Developed an AI pipeline using fine-tuned **Gemma** model to generate adaptive learning content.

Human-Computer Interaction Researcher | Vibe AI

Human-AI Interaction and User Research

- Conducted generative **user research** using **focus groups, semi-structured interviews, and surveys** to investigate user needs, expectations, pain points, and opportunity areas for an AI Product
- Investigated how users interacted with AI product concepts and identified recurring **behavioral patterns**, interaction barriers, decision triggers, and sources of UX friction across the user journey.
- Evaluated multiple product prototypes with 100+ users, combining usability metrics with post-use interviews to identify recurring interaction problems and behaviour for actionable design insights.
- Synthesized qualitative and **behavioral evidence** into product recommendations and collaborated with the development team to translate research findings into human-centered improvements to AI interactions.

Research Project

Zomato Consumer and Usability Research

- Conducted **behavioral and consumer research** examining how users discover, evaluate, compare, and book restaurants, with a focus on behavioral patterns, and friction within the digital service journey.
- Conducted task-based **behavioral observation** to identify navigation patterns, errors, information-seeking behavior, decision triggers, and points of interaction friction across the user journey
- Translated observed **behavioral patterns** into redesigned interface concepts and evaluated alternative prototypes through **usability testing** and **A/B testing**
- Demonstrated an 18% reduction in **task-completion time** and 15% increase in task-completion rate, providing quantitative evidence for the effectiveness of behavior-informed design changes.

Research Intern, Department of Psychology, King's College London (remote collaboration)

Bayesian Computational Modelling of Pain Perception. Supervised by Dr. Devin Terhune

- Conducted advanced **Bayesian computational modelling** to re-analyse placebo hypoalgesia data on a published study's experimental data, performing exploratory statistical analysis using state

space models, yielding novel insights and generating new research questions for further analyses.

- Systematically applied and compared multiple competing computational models to identify the best-fitting account of pain perception, achieving **84% predictive** accuracy; utilised **MATLAB's Variational Bayesian Analysis (VBA) toolbox** for model inversion and parameter estimation.
- Drafted the **study design** for conducting **confirmatory analyses** based on the exploratory analyses performed. Drafted the **research question**, **devised the hypotheses**, and ideated a comprehensive **analyses plan** for hypotheses testing.
- Drafted comprehensive pre-registration document to pre-register the study on **Open Science Forum (OSF)** [[Link to the OSF Registration](#)]

Working Papers

- Mishra, U., Parsons R., Wiech K. & Terhune, D. (*in preparation*). Investigating the link between trait responsiveness to verbal suggestion and the placebo effect within the predictive processing framework. *Preprint in preparation for journal submission.*

Conference Selections

- Mishra, U., Parsons R., Wiech K. & Terhune, D. (2026). Investigating the link between trait responsiveness to verbal suggestion and the placebo effect within the predictive processing framework. *Accepted for Poster Presentation at the 13th Annual Conference of Cognitive Science (ACCS).* (Unable to attend.)
- Mishra, U. & Kosch, T. (2026). Impact of Perceived AI-based Assistance on User Experience and Task Performance. *Accepted for Poster Presentation at the 13th Annual Conference of Cognitive Science (ACCS).* (Unable to Attend.)

Skills

- **Statistical software:** R (proficient), Python (proficient: pandas, numpy, scipy, sklearn), MATLAB
- **Research methods:** Survey design, factor analysis (EFA/CFA), regression analysis, Human-AI Research, sentiment analysis, causal inference methods, Bayesian statistical modelling, Mixed Method Research, scientific reporting
- **Data collection tools:** Qualtrics, PsychoPy, Eye Tracking, Keyboard Clicks
- **AI / NLP:** Multimodal BERT, Natural Language Processing, ensemble learning, Variational Bayesian Inference, Bayesian Modelling
- **Other tools:** Git/GitHub, LaTeX, MS Excel

Languages

- **Hindi:** Native proficiency
- **English:** Full professional proficiency

Referees

Prof. Jeevant Rampal

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Indian Institute of Management Ahmedabad.
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Prof. Anindya Chakrabarti

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